

Yousuf Rashid

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EDUCATION

University of Waterloo

Sep 2025 – Present

Bachelor of Mathematics (Honours)

Waterloo, ON

- in Applied Mathematics with Scientific Computing and Scientific Machine Learning

EXPERIENCE

Research Engineering Intern

Apr 2026 – Aug 2026

DeepIDV

San Francisco, CA

- Co-designed the algorithm for a three-party consent-gated decryption protocol requiring collaborative recovery across Client, HSM-backed Operator, and Relying Party (**Patent Pending**)
- Eliminated hashing failures from biometric variance by removing ArcFace embeddings from the encryption boundary and introducing **fuzzy extraction**
- Generated 2,500+ synthetic IDs across 100+ document types with YOLO and PIL, powering a TensorFlow classifier that reached **99.8%** accuracy

Machine Learning Researcher

Jan 2026 – Present

Wat.AI

Waterloo, ON

- Diagnosed FinBERT's macro-headline performance as a **sentiment-inversion bug** rather than a data issue
- Fine-tuned [finbert-macro](#) with Hugging Face Transformers, unfreezing the top 6 encoder blocks to gain **+29.8** macro F1 without a full retrain
- Replaced 113 hand-labeled headlines with market-reaction distant supervision, validated via purged walk-forward CV with a 7-day embargo

PROJECTS

Constrained Multi-Agent RL for Spatial Control Tasks (Soccer Based Environments)

[GitHub](#)

- Rewrote a spatial-control model as a **custom CUDA kernel in C++**, delivering a **5.44x** throughput gain at 0.0009 max deviation from a known NumPy implementation
- Implemented Lagrangian-constrained PPO (Roy et al., 2022) in PyTorch over a 10-agent MultiDiscrete action space, outperforming a 10M-step unconstrained PPO baseline after just **2.5M** steps
- Calibrated a correlated-Gaussian generative model of multi-agent formations against real-world tracking data
- Found patching an emergent side effect in Lagrangian-constrained PPO cost **15pp** of task success, isolating a real trade-off in multi-constraint RL

Sequential Ranking Policy with Counterfactual Evaluation (NBA Fantasy Drafts)

[GitHub](#)

- Built a **two-stage ranking** system with a **XGBoost** candidate scoring into a **Monte Carlo** rollout policy
- Modeled candidate availability with a **Cox survival** model, reaching **0.935** concordance
- Ran counterfactual evaluation over **4,400** paired drafts, controlling for season/league size/slot/seed
- Rejected a **+2.35** pt lift as noise via confidence interval; underperformed ADP baseline by **-0.14** pts

Overfit Detection on Trading Strategies

[GitHub](#)

- Built distribution diagnostics using **bootstrap sampling** and **Monte Carlo** simulation to detect p-hacked and overfit trading strategies, flagging false positives with **68%** accuracy and avoiding \$2,000 in paper-trading losses
- Optimized the simulation engine to support **10,000+** simulations per query, estimating price-change distributions conditioned on technical indicators (RSI, moving averages, volatility)
- Developed a specific query layer with regex parsing, translating SQL-inspired syntax into executable logic

TECHNICAL SKILLS

Languages: Python, C++, C, SQL

Data Analysis: pandas, NumPy, SciPy, matplotlib

Modeling: scikit-learn, XGBoost, PyTorch, Tensorflow, Transformers (Hugging Face)

Frameworks: Git, CUDA, PostgreSQL, FastAPI, SQLite, Linux